

Data Science

COMP5122M

What is Data Science?

Duygu Sarikaya

School of Computing, University of Leeds

Lecturers



- Duygu Sarikaya, Module Leader



- Paula Satne, Ethics

- Guest Lecture / Meet a Data Scientist

Teaching Assistant

Marking Support:

Dr. Shabbar Naqvi

Dr. Zeeshan Babar

Dr. Noorhan Abbas

Practicals/Lab Support:

Abhimanyu Gaur, Arpita Saggar, Oliver Mills, Raneem Toman, Edward Ellis, Xin Ci Wong, George Baker, Mohsen Azarmi

- Session Allocations To Be Announced (TBA)

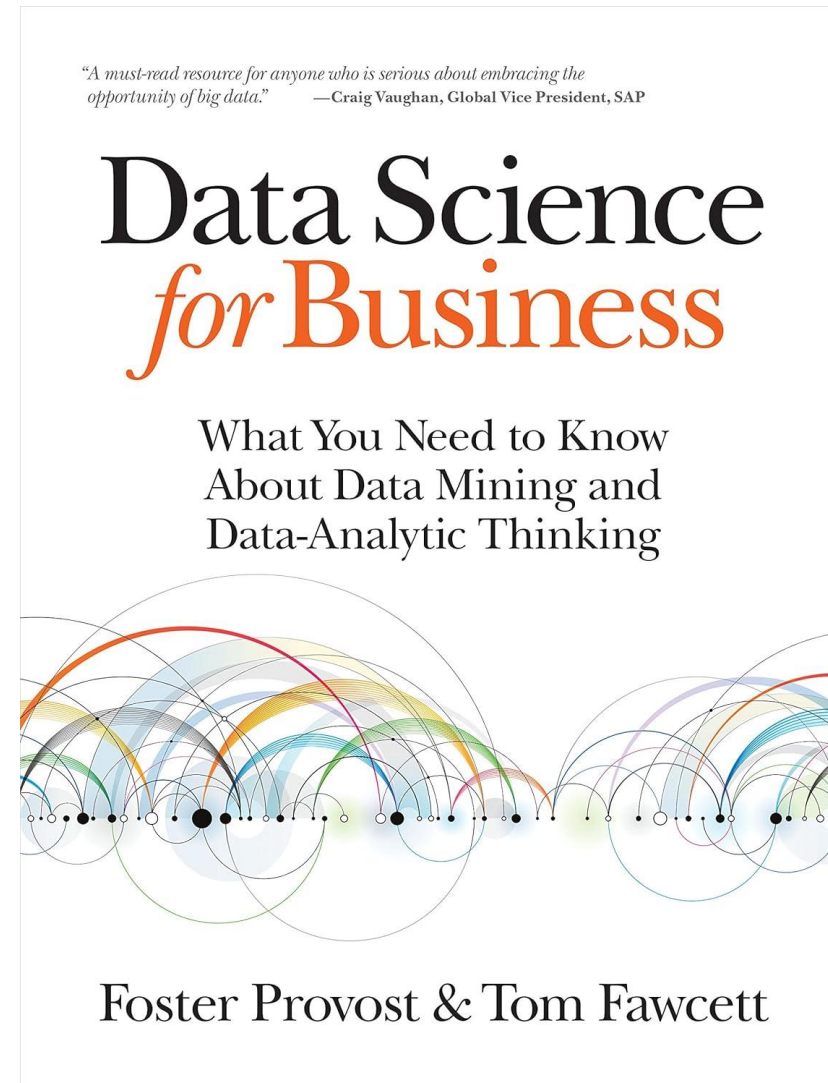
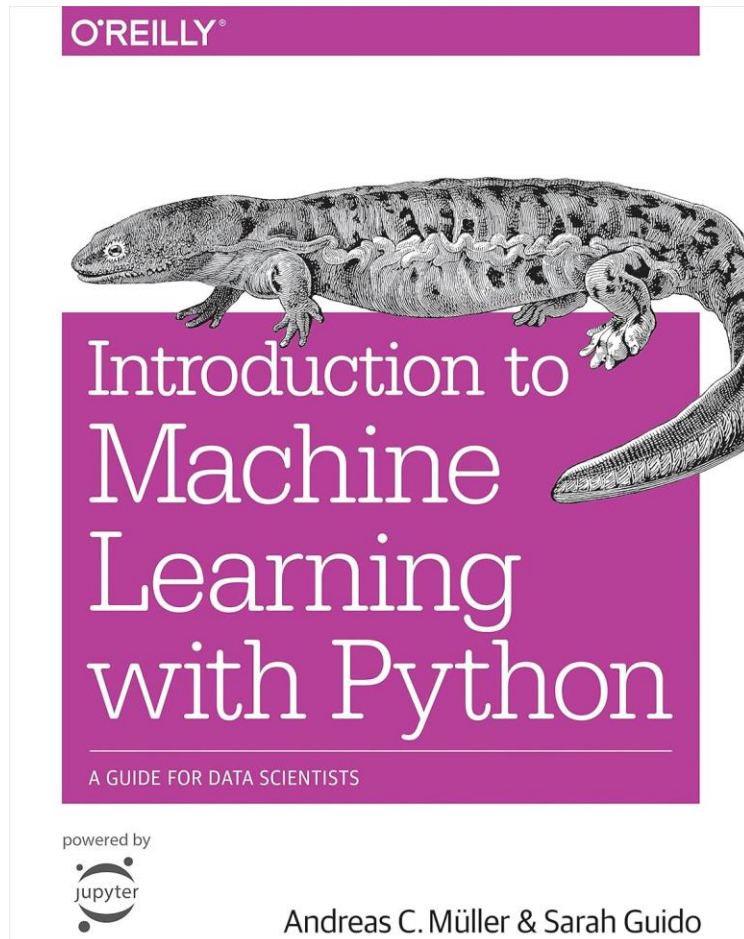
Location/Time

- Please Check your Timetables!
- Lectures (Wednesday/Friday)
 - Wednesday 15:00 - 16:00 at Conference Auditorium 1 (GM.03)
 - Friday 12:00 – 13:00 at Esther Simpson LT (LG.08)
- Practicals/Labs (3 of them in total)
 - Sessions on Wednesdays and Fridays (Check your timetable)
- Term Dates
 - Monday 28 September to Friday 12 December 2025
 - <https://www.leeds.ac.uk/term-dates>
- Examinations
- **Autumn semester:** Monday 12 January to Friday 23 January 2026

Materials

- Aim to make slides available at least 48 hours before the lecture
- However, I may continue to update them, the final version will be the one that is available by midnight on the day the lecture is completed
- Tutorials and code examples will be added either before or after the lectures
- Sample answers to practicals will be shared only after all the sessions are completed and the last session is given enough time to complete the assignment

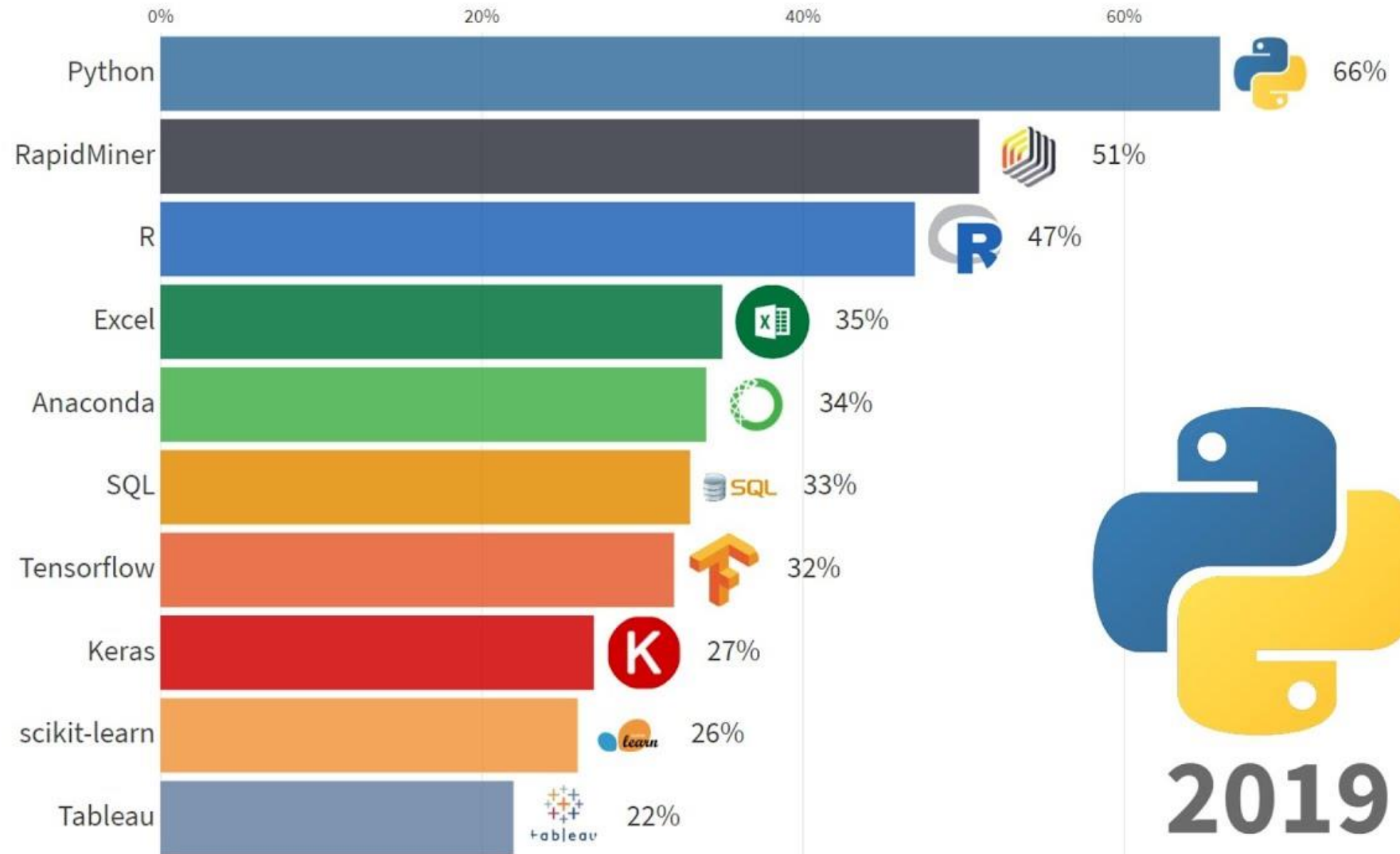
Materials (Readings, Minerva>Reading List for full list)



Module Delivery Plan (Updated)

Week Number	Teaching topic		Assessment
1 (29/9/25)	What is Data Science?	Data Quality	
2 (6/10/25)	Exploratory Data Analysis and Visualization	Exploratory Data Analysis and Visualization	Practical 1 (formative)
3 (13/10/25)	Machine Learning Overview	Data Similarity and Clustering	
4 (20/10/25)	Classification	Decision Trees	Practical 2 (formative) / Groups assigned
5 (27/10/25)	Reading Week	Reading Week	Coursework Part I Out (summative)
6 (3/11/25)	Regression	Model Fitting	
7 (10/11/25)	Model Evaluation	Dimensionality Reduction (Singular Value Decomposition)	Coursework Part I (summative) (deadline 14 th of November)
8 (17/11/25)	Principal Component Analysis	Case Study/Evidence of Fraud – a Detective Story	Practical 3 (formative)
8 (24/11/25)	Revision / Guest lecture by LIDA (Meet a Data Scientist)	Ethics - Introduction to Ethics and Data Governance	Coursework Part II Out (summative)
9 (1/12/25)	Ethics - Introduction to Ethics and Data Governance	Ethics - Privacy and Bias	
10 (8/12/25)	Ethics - Privacy and Bias	Revision	Coursework Part II (summative) (deadline 12 th of December)

Practicals/Labs



Assessment

- Case Study / Coursework (%50)
In groups with max. 4 people
- Final Exam (%50)
 - Written Exam
 - On Campus

Module Success Plan: Choose Your Setting

Beginner



- Attend Lectures
- Attend Practicals/labs and complete the assignments
- Read Relevant Chapters from Recommended Books
- Follow tutorials, code examples /complete them on your own



Module Success Plan: Choose Your Setting

Intermediate



- Work on side projects with real-life datasets
- Go above and beyond with visualization techniques
- Good insights and storytelling



Module Success Plan: Choose Your Setting

Expert



- Complete at least one major project that involves web scraping and data curation (God tier)
- Go above and beyond with visualization and storytelling
- Read state-of-the-art research papers





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What is Data Science?

Join at:
vevox.app

ID:
187-603-324



<https://universityofleeds.vevox.com/#/meeting/791814/polls>

What is Data Science?

- The purpose of computing is insight, not numbers.
- – Richard W. Hamming



What is Data Science?

Kenneth Neil Cukier and Viktor Mayer-Schoenberger wrote an article called “The Rise of Big Data”. In it, they discuss the concept of datafication

- They define datafication as a process of “taking all aspects of life and turning them into data.”

What is Data Science?

- Example: How we quantify “health and wellbeing” using a smartwatch
- Take someone who wears a fitness tracker. Without really thinking about it, parts of their daily life such as sleep, steps, heart rate, and movement are turned into data they can see.
- Over time, that data helps them notice things they would’ve missed:

What is Data Science?

- Keep feeling tired in the mornings even when they thought they had slept enough.
- When they check their sleep data, they notice they were waking up several times a night without realising. That helped them connect the problem to late-night screen time and caffeine.

What is Data Science?

- Their watch also tracks their resting heart rate.
- One month, they saw it slowly creeping up. They hadn't noticed any changes in how they felt, but that little number made them pay attention. They realised they had been working longer hours and skipping their evening walks.

What is Data Science?

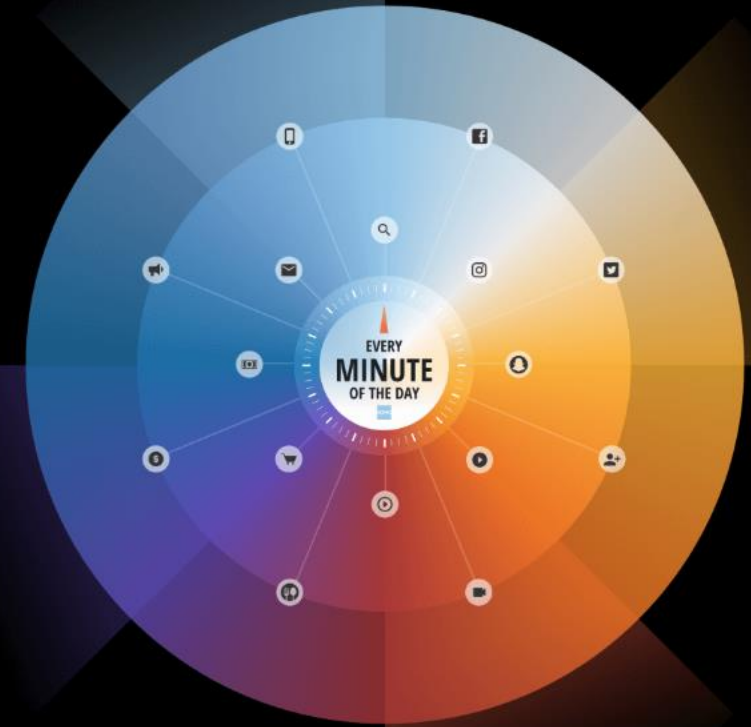
- Once we datafy things, we can transform their purpose and turn the information into new forms of value.
- The app shows gentle reminders and weekly trends. When they sit for too long, it nudges them to move.
- When they go to their GP about feeling low-energy and stressed, they didn't just have to describe things vaguely. They could show the doctor their sleep patterns and heart-rate trends over a few months. That helped the doctor give more tailored advice, instead of guessing.

What is Data Science?

Data Never Sleeps 10.0

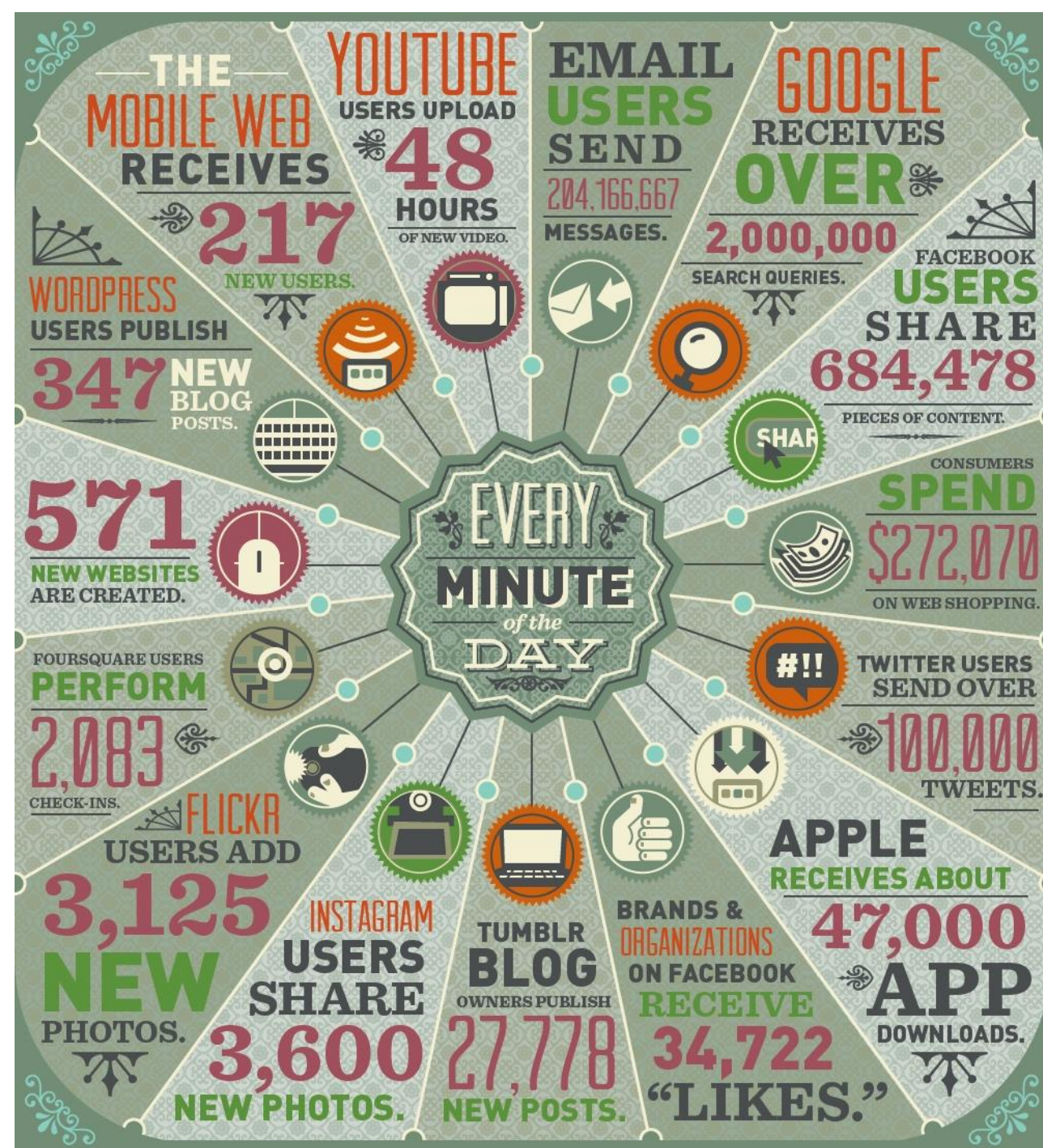
In this 10th annual "Data Never Sleeps" infographic, we share a glimpse at the data being created each minute on the internet. It's a breathtaking view on how the volume and variety of data keeps accelerating, showing no signs of slowing down.

[VIEW THE INFOGRAPHIC](#)



How much data is
generated ***every minute?***

Version 1.0 (2013)

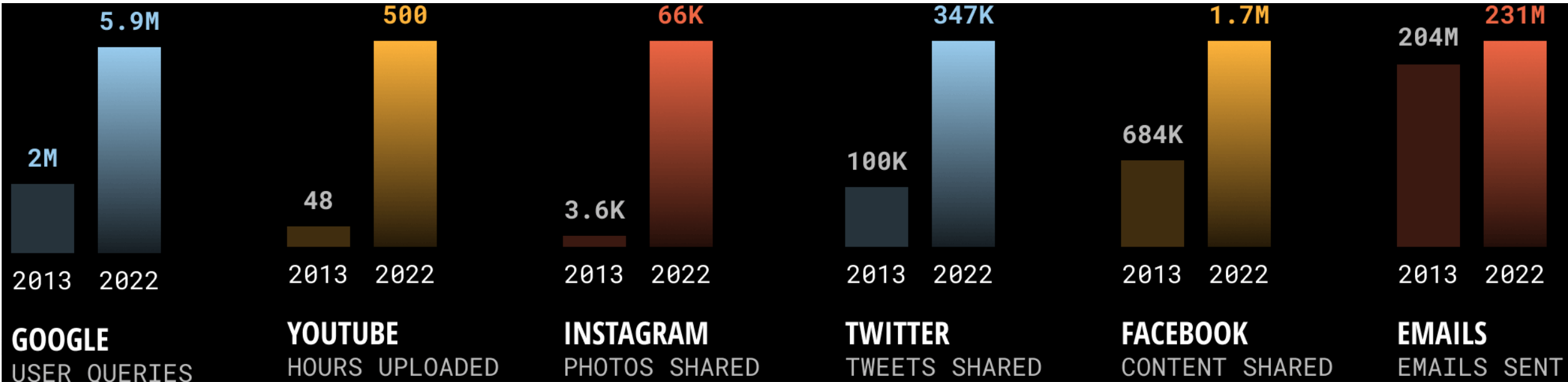


How much data is generated *every minute*?

Version 10.0 (2022)

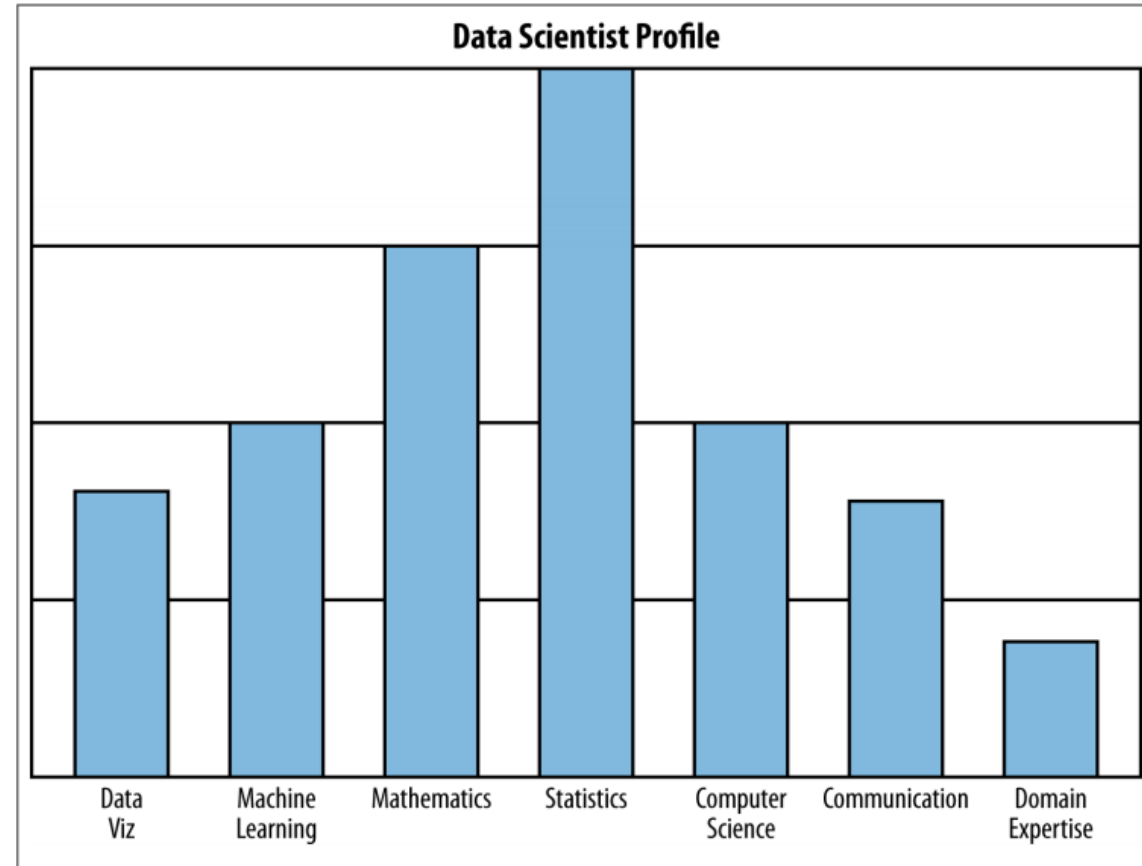
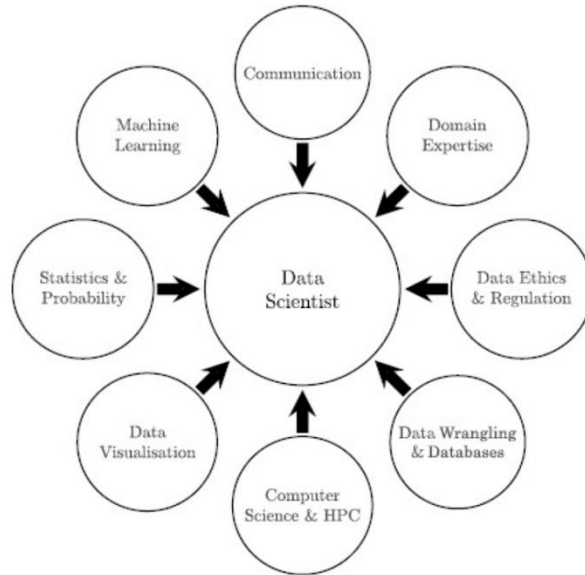


Data keeps growing with no signs of slowing

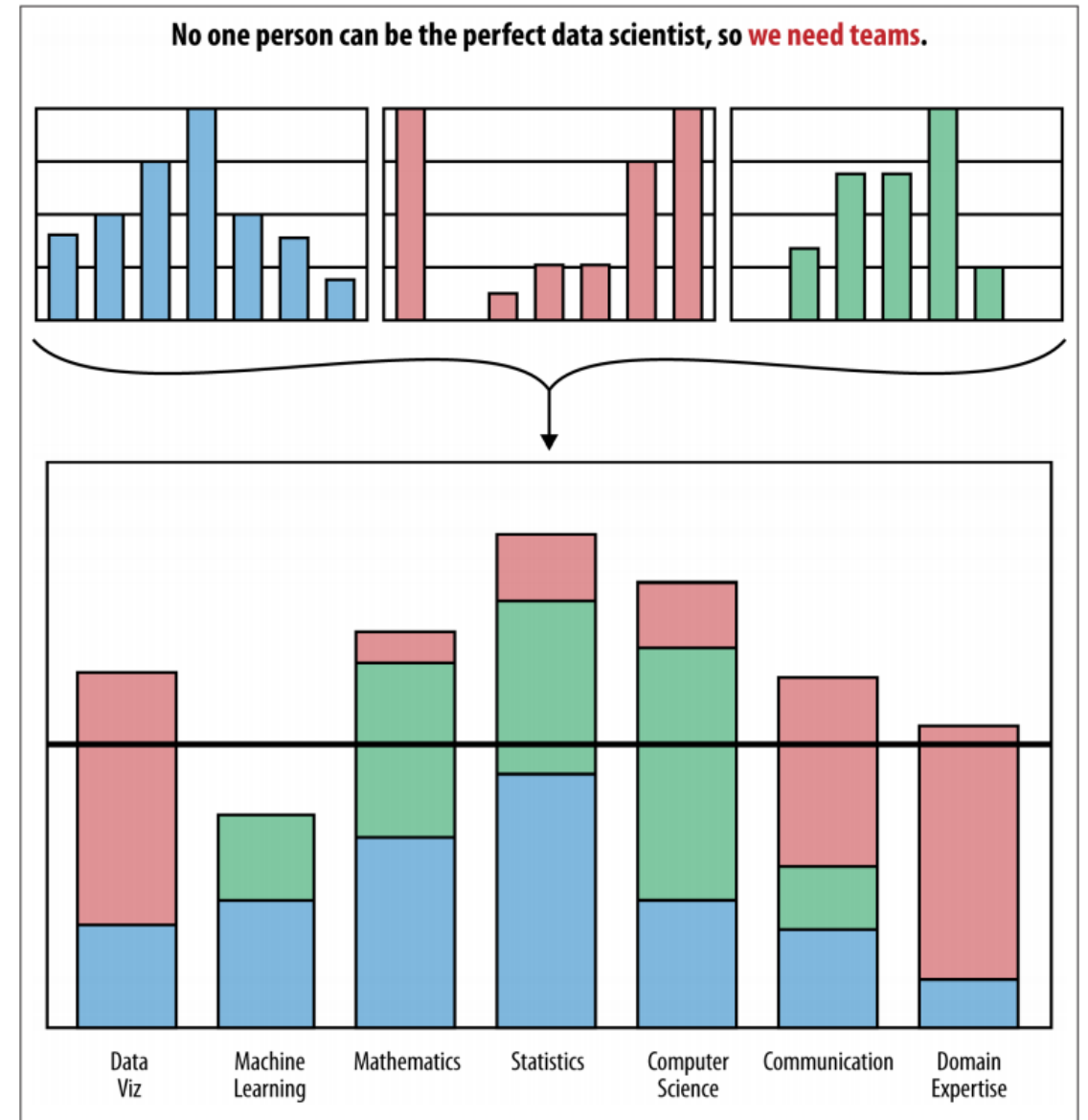


Profile of a Data Scientist

- Computer science
- Math
- Statistics
- Machine learning
- Domain expertise
- Communication and presentation skills
- Data visualization



Profile of a Data Scientist



The Internet Movie Database (IMDb)

- Crowdsourced and curated data about all aspects of the motion picture industry, at www.imdb.com
- Data on over 3.3 million movies and TV programs

+

Forrest Gump (1994)

7+ | 2h 22min | Drama, Romance | 11 November 1994 (Turkey)

★ 8,8

1.767.629

☆ Rate

★ This



3:54 | Trailer

11 VIDEOS | 233 IMAGE

The presidencies of Kennedy and Johnson, the events of Vietnam, Watergate and other historical events unfold through the perspective of an Alabama man with an IQ of 75, whose only desire is to be reunited with his childhood sweetheart.

Director: Robert Zemeckis

Writers: Winston Groom (novel), Eric Roth (screenplay)

Stars: Tom Hanks, Robin Wright, Gary Sinise | [See full cast & crew »](#)

▶

Watch on Prime Video

included with Prime

+

Add to Watchlist

82

Metascore

From metacritic.com

Reviews

2.449 user | 169 critic

Popularity

219 (♦ 28)

The Internet Movie Database (IMDb)

For each film, IMDb includes:

- its title, running time, genres, date of release, and a full list of cast and crew.
- financial data about each production, including the budget for making the film and how well it did at the box office.
- ratings for each film from viewers and critics (scores on a zero to ten stars scale)
- written reviews
- links between films: for example, identifying which other films have been watched most often by viewers of a film

The Internet Movie Database (IMDb)

Details

[Edit](#)

Official Sites: [Official Facebook](#)

Country: [USA](#)

Language: [English](#)

Release Date: 11 November 1994 (Turkey) [See more »](#)

Also Known As: [Forrest Gump](#) [See more »](#)

Filming Locations: [Monument Valley, Arizona, USA](#) [See more »](#)

Box Office

[Edit](#)

Budget: \$55,000,000 (estimated)

Opening Weekend USA: \$24,450,602, 10 July 1994

Gross USA: \$330,455,270

Cumulative Worldwide Gross: \$678,226,805

[See more on IMDbPro »](#)

Company Credits

Production Co: [Paramount Pictures](#) [See more »](#)

[Show more on IMDbPro »](#)

Technical Specs

Runtime: 142 min

Sound Mix: [DTS \(The Digital Experience\)](#) | [Dolby SR](#) | [SDDS](#) | [Dolby Atmos](#) | [Dolby Digital \(5.1\)](#)

Color: [Black and White](#) | [Color](#)

Aspect Ratio: 2.39 : 1

[See full technical specs »](#)

Storyline

[Edit](#)

Forrest Gump is a simple man with a low I.Q. but good intentions. He is running through childhood with his best and only friend Jenny. His 'mama' teaches him the ways of life and leaves him to choose his destiny. Forrest joins the army for service in Vietnam, finding new friends called Dan and Bubba, he wins medals, creates a famous shrimp fishing fleet, inspires people to jog, starts a ping-pong craze, creates the smiley, writes bumper stickers and songs, donates to people and meets the president several times. However, this is all irrelevant to Forrest who can only think of his childhood sweetheart Jenny Curran, who has messed up her life. Although in the end all he wants to prove is that anyone can love anyone.

Written by [aliw135](#)

[Plot Summary](#) | [Plot Synopsis](#)

Plot Keywords: [vietnam war](#) | [amputee](#) | [war hero](#) | [based on book](#) | [based on novel](#) | [See All \(264\) »](#)

Taglines: Life is like a box of chocolates...you never know what you're gonna get. [See more »](#)

Genres: [Drama](#) | [Romance](#)

Certificate: 7+ | [See all certifications »](#)

Parents Guide: [View content advisory »](#)

The Internet Movie Database (IMDb)

Tom Hanks

Producer | Actor | Soundtrack



1:58 | Trailer

202 VIDEOS | 1658 IMAGES

Thomas Jeffrey Hanks was born in Concord, California, to Janet Marylyn (Frager), a hospital worker, and Amos Mefford Hanks, an itinerant cook. His mother's family, originally surnamed "Fraga", was entirely Portuguese, while his father was of mostly English ancestry. Tom grew up in what he has called a "fractured" family. He moved around a great ... [See full bio »](#)

Born: July 9, 1956 in Concord, California, USA

[More at IMDbPro »](#)

[Contact Info:](#) View agent, publicist, legal on IMDbPro

Top 500

Filmography

Show all | Show by... | Edit

Jump to: [Producer](#) | [Actor](#) | [Soundtrack](#) | [Writer](#) | [Director](#) | [Thanks](#) | [Self](#) | [Archive footage](#)

Producer (58 credits)

[Hide](#)

[A Man Called Ove](#) (producer) (*announced*)

[Beautiful](#) (producer) (*announced*)

[No Better Place to Die](#) (executive producer) (*announced*)

[In the Garden of Beasts](#) (producer) (*pre-production*)

[Masters of the Air](#) (TV Mini-Series) (executive producer - 9 episodes) (*announced*)

- Episode #1.9 ... (executive producer)
- Episode #1.8 ... (executive producer)
- Episode #1.7 ... (executive producer)
- Episode #1.6 ... (executive producer)
- Episode #1.5 ... (executive producer)

[Show all 9 episodes](#)

[The Movies](#) (TV Series documentary) (executive producer - 6 episodes) 2019

- The Golden Age (2019) ... (executive producer)
- The Sixties (2019) ... (executive producer)
- The Seventies (2019) ... (executive producer)
- The 2000s to Today (2019) ... (executive producer)
- The Nineties (2019) ... (executive producer)

[Show all 6 episodes](#)

[The 2019 Rock and Roll Hall of Fame Induction Ceremony](#) (TV Special) 2019 (executive producer)

[The 2000s](#) (TV Series documentary) (executive producer - 7 episodes) 2018

- I Want My MP3 (2018) ... (executive producer)
- Yes We Can (2018) ... (executive producer)
- The Financial Crisis (2018) ... (executive producer)
- The i Decade (2018) ... (executive producer)

961884/4 [mire: Bush's Second Term](#) (2018) ... (executive producer)

The Internet Movie Database (IMDb)

What kind of questions can you answer with this data?

Questions



Data

The Internet Movie Database (IMDb)

What kind of questions can you answer with this data?

Questions



Data



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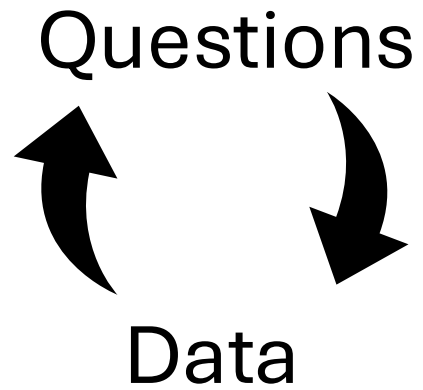
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182-934-583



<https://universityofleeds.vevox.com/#/meeting/791820/polls>

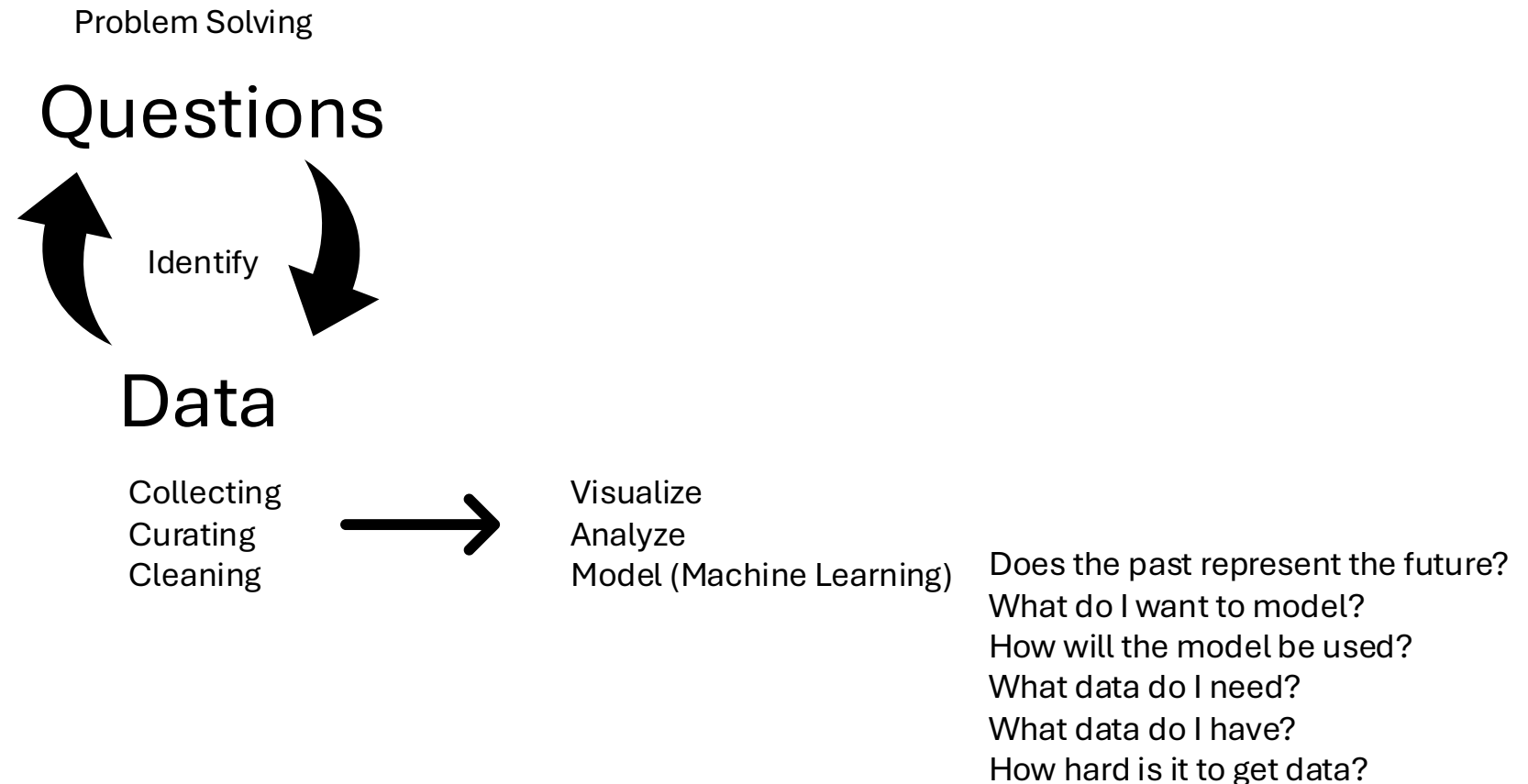
The Internet Movie Database (IMDb)

What kind of questions can you answer with this data?



The Internet Movie Database (IMDb)

What kind of questions can you answer with this data?



The Internet Movie Database (IMDb)

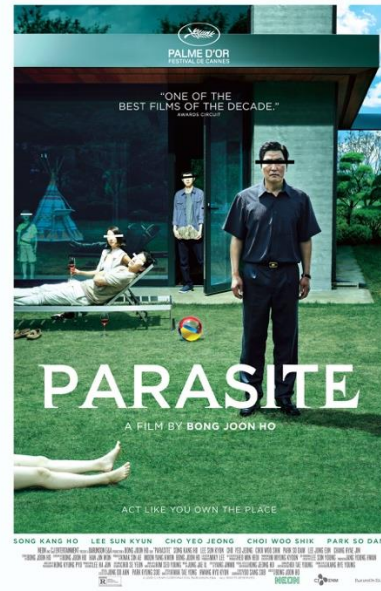
What kind of questions can you answer with this data?

- Which actors appeared in the most films? Appeared in the lowest rated films? Had the longest career or the shortest lifespan?
- What was the highest rated film each year, or the best in each genre? Which movies lost the most money, had the highest-powered casts, or got the least favorable reviews.

The Internet Movie Database (IMDb)

What kind of questions can you answer with this data?

- How well does movie gross correlate with viewer ratings or awards?
- How do Hollywood movies compare to Korean movies, in terms of ratings, budget, and gross? Are foreign movies better received than American films, and how does this differ between U.S. and non-U.S. reviewers?



The Internet Movie Database (IMDb)

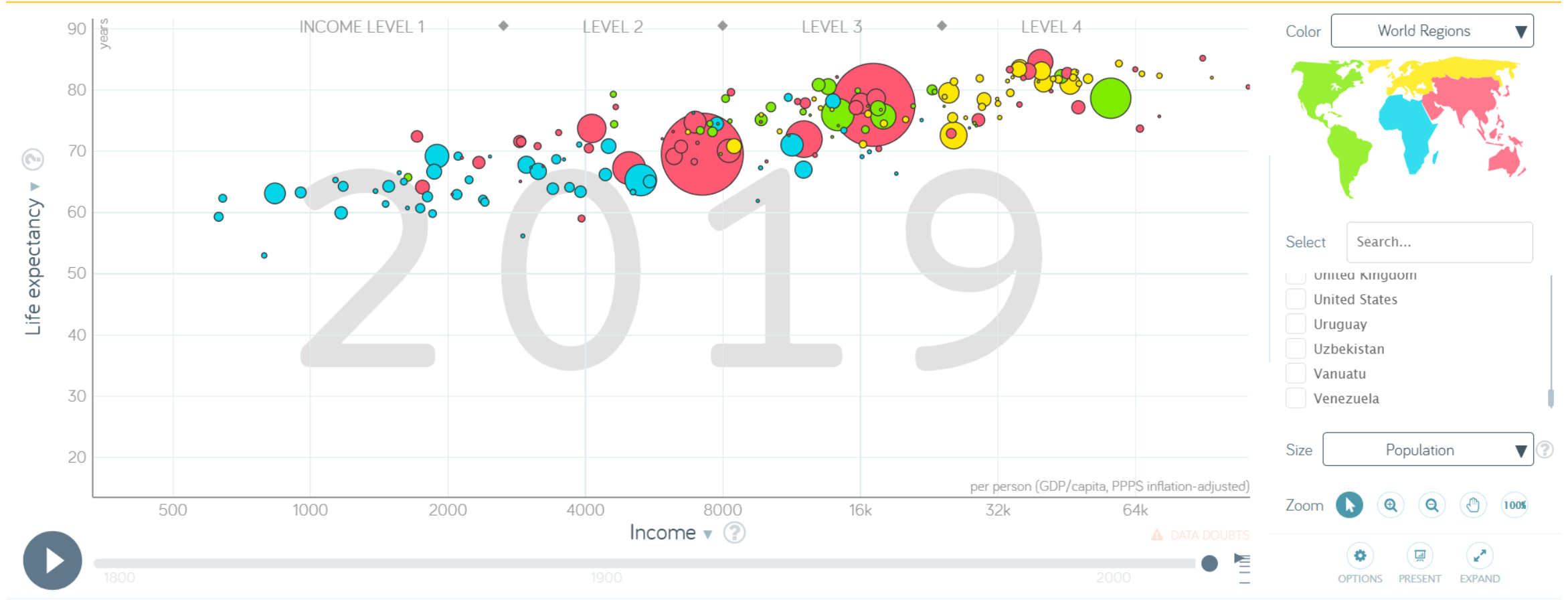
What kind of questions can you answer with this data?

- What is the age distribution of actors and actresses in films?
- How much younger is the actress playing the wife, on average, than the actor playing the husband?
- Has this disparity been increasing or decreasing with time?
- In which movies did Tom Hanks played with Keanu Reeves?



Gapminder Wealth and Health of Nations

- <https://www.gapminder.org/answers/how-does-income-relate-to-life-expectancy/>
- [https://www.gapminder.org/tools/?from=world#\\$state\\$time\\$value=2019;;&chart-type=bubbles](https://www.gapminder.org/tools/?from=world#$state$time$value=2019;;&chart-type=bubbles)



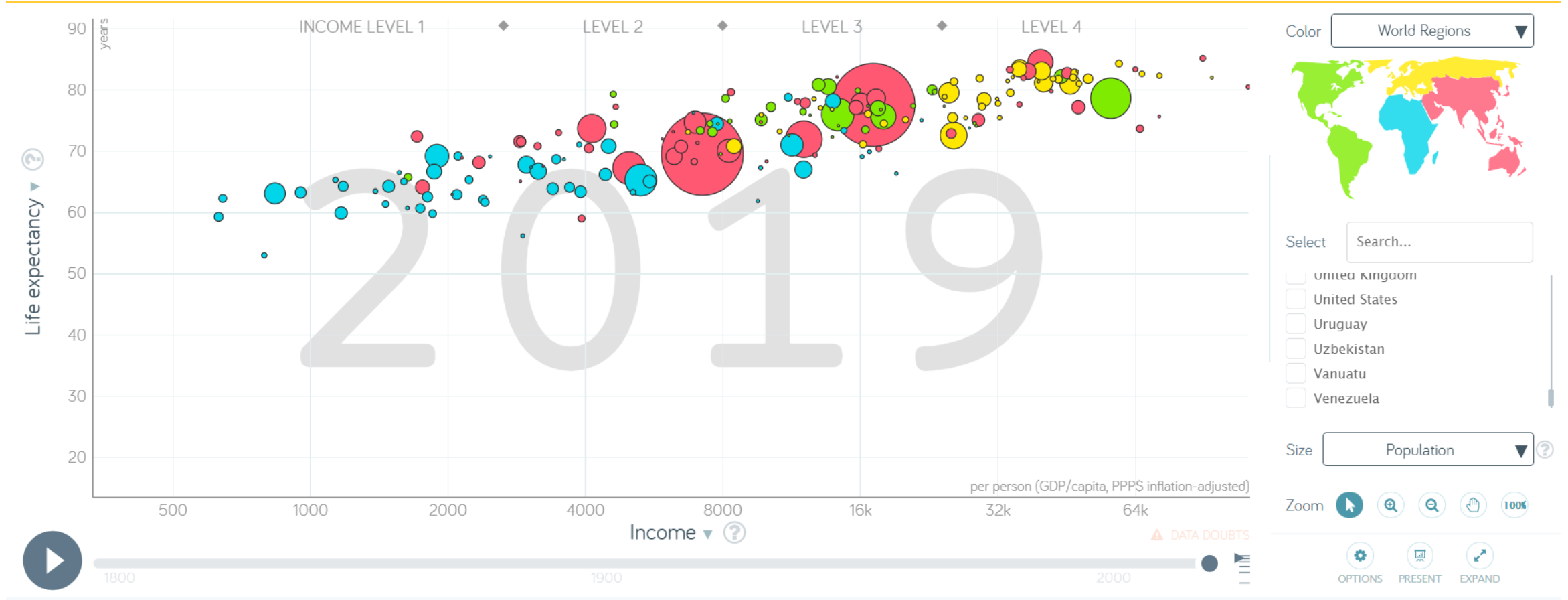
Gapminder Wealth and Health of Nations

GDP per capita (Gross Domestic Product per person), which reflects a country's total economic output divided by its population.

- An indicator of a nation's wealth
- Useful for **cross-country comparisons** of economic resources and their effects on outcomes like life expectancy

Gapminder Wealth and Health of Nations

- <https://www.gapminder.org/answers/how-does-income-relate-to-life-expectancy/>
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Objectives

- The heart of data science lies in **doing the simple things right**: understanding the application domain, cleaning and integrating relevant data sources, and presenting your results clearly to others.
- It takes considerable insight and experience to **ask the right questions**, and sense whether you are moving toward correct answers and **actionable insights**.

What is Data Science?

- Data science encompasses a set of principles, problem definitions, algorithms, and processes for extracting nonobvious and useful patterns from large data sets.
- Machine learning (ML) focuses on the design and evaluation of algorithms for extracting patterns from data.
- Data mining generally deals with the analysis of structured data and data discovery
- Data science also takes up other challenges, such as the capturing, cleaning, and transforming of unstructured social media and web data; the use of big-data technologies to store and process big, unstructured data sets; and questions related to data ethics and regulation.

What is Data Science?

- Data science rests on a foundation of mathematics, particularly statistics and linear algebra. It is important to understand this material on an intuitive level: why these concepts were developed, how they are useful, and when they work best.
- Approaches which come most naturally to computer scientists, particularly the algorithmic manipulation of data, the use of machine learning, and the mastery of scale
- Core values of statistical reasoning: the need to understand the application domain, proper appreciation of the small, the quest for significance, and a hunger for exploration.

TODO (This week):

Reading Assignment

50 years of Data Science by David Donoho

TODO (Now):

Introduce yourself to at least two people you have never met before

- Hello, my name is ...
- I'm from ...
- I want to do this module/degree because
- In my spare time I like to ...

